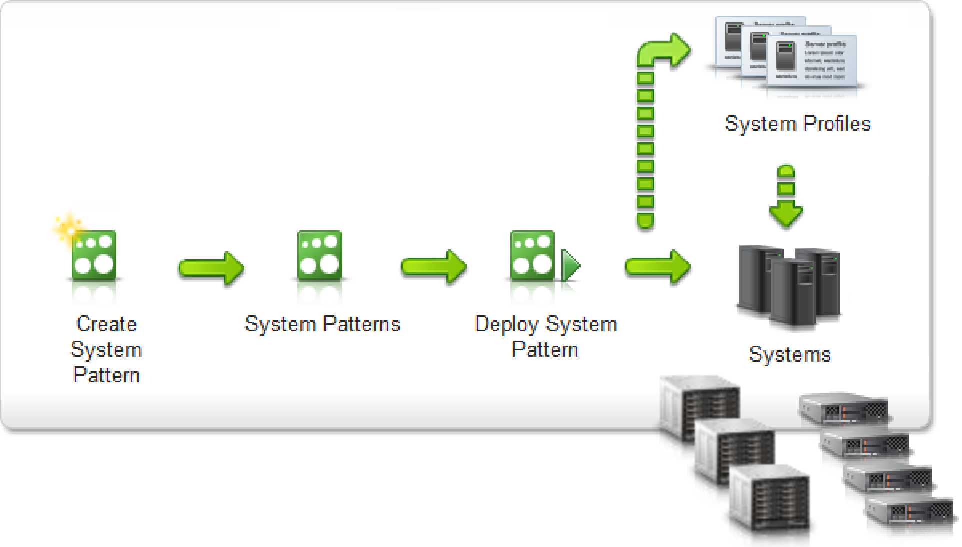
LXCA Configuration Patterns

David Roberts | Configuration Patterns– 4/15/2015

Configuration Patterns Introduction

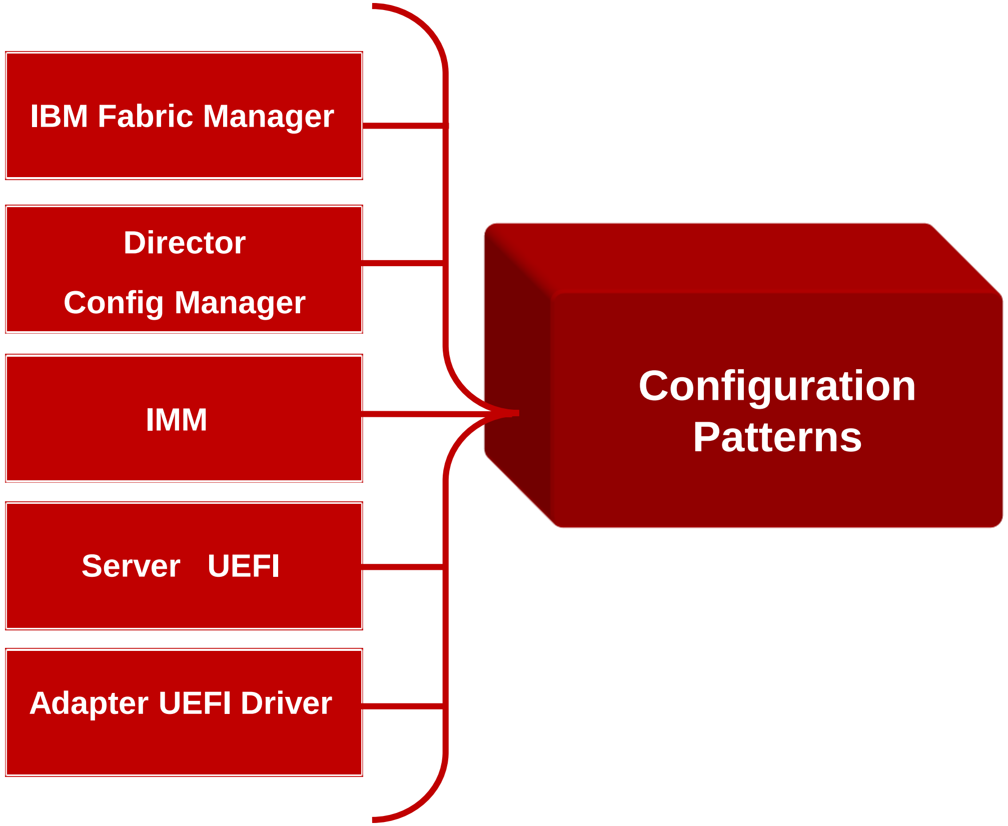
An integrated configuration management solution that simplifies the end-to-end task of provisioning System x hardware resources



Client Value

- Reduce time spent on hardware rollout
  - Deploy pre-OS configuration to multiple devices from a single pattern
  - Integrate device configuration into single interface and workflow
    local storage, IMM, UEFI, I/O adapter and VNIC settings, virtual MAC/WWN addressing, SAN boot target
- Centralized pattern based management of configuration lifecycle
  - Consistent configuration baseline
  - Change configuration of multiple devices from a single interface
- Enable support for pre-configuration of ordered hardware, before physical acquisition
  - Pre-provision hardware by deploying placeholder configurations

Configuration Patterns: Integrated Solution



- Consolidate operations into 1 unified solution that provides 1-to-many automation
- Guided approach to creating pools, patterns, and profiles
- Create reference patterns by capturing from existing hardware
- Tight binding of configurations to managed devices
- Propagation of pattern configuration changes
- Assistance with device/pattern differences

Configuration Patterns: Differences from IBM FSM version

- Chassis Pattern support not available
  - Chassis component (Nodes, CMM, IOMs, etc) network configuration available Chassis Component Network Configuration UI

- Compute node failover support not available
  - Manual failover supported via profile deactivate / reactivate

- In order to clarify confusion related to server profile actions, some terminology changes have been implemented
  - Deploy profile is now called “Activate Profile”
  - Unassign profile is now called “De-activate Profile”
  - Profile status states are now ‘Active’ and ‘Inactive’
  - Deploy Pattern action remains unchanged

Configuration Patterns: New Features

- User defined Ethernet & Fibre address pools
  - User supplied definition of address ranges
- Virtual address usage reporting
  - Expand support to include a view of the active ports and assigned addresses
- Pattern compatibility confirmation before deployment
  - Provide user with ability to review and confirm compatibility with servers before proceeding with deployment
- Parallel Deployment
  - Process queue in parallel, improving deploy processing time
- Pre-defined UEFI Patterns
  - Provide client with predefined UEFI patterns that match ATS recommendations for common workloads (Low Latency, Virtualization, etc)
- Import & Export of Patterns
  - REST API support for importing and exporting server pattern and associated category patterns

Configuration Patterns: Hardware Enhancements

- Broaden compute node configuration extending support to rack based systems
  – Integrate rack servers with Flex node workflow, avoid dissimilar experiences
  – Support RAID, I/O Adapter, & IMM/UEFI Settings
  – Address virtualization not supported

- Supported Servers:
  – x3550 M5, x3650 M5, x3960 X6

- Provide support for key chassis switch configuration
  – Add key switch configuration functions for Lenovo Flex chassis Ethernet switches
  – Focused on functions that require coordination of ITE level and Chassis switch configuration
    – eg. vNIC/UFP & VLAN ID tags

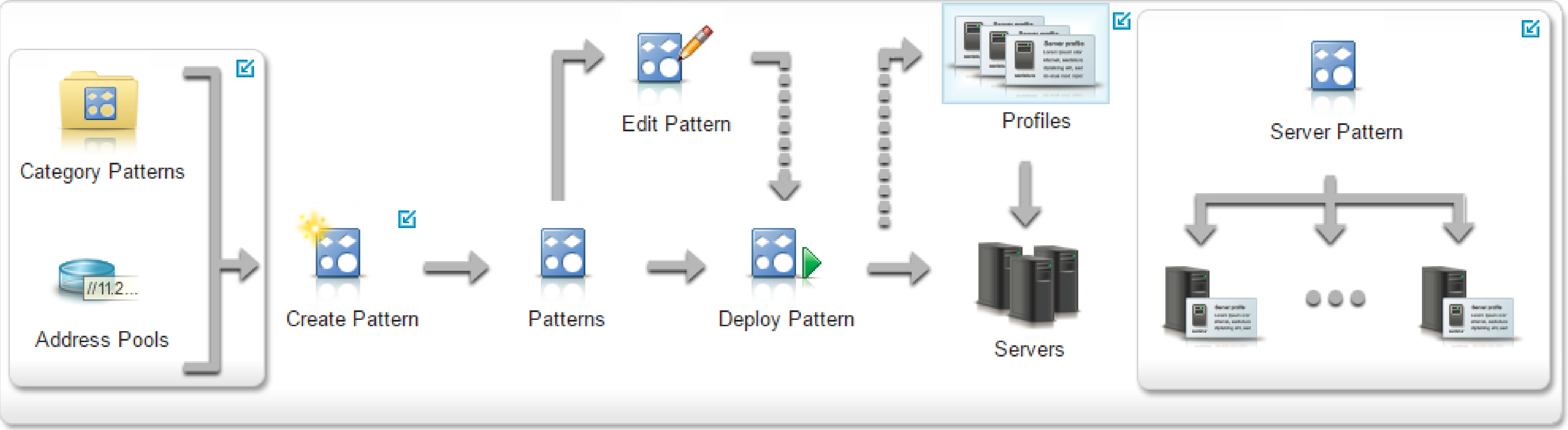
Configuration Patterns: Getting Started

- Introduction to concepts, including user objects and workflow

Suggested Tasks (1) | Concept Overview

Using Configuration Patterns, you can quickly provision or pre-provision multiple servers from a single pattern. Subsequent pattern changes are automatically applied to all associated servers. To learn more, hover over an icon in the diagram, which depicts the high-level objects and workflow for using Configuration Patterns.

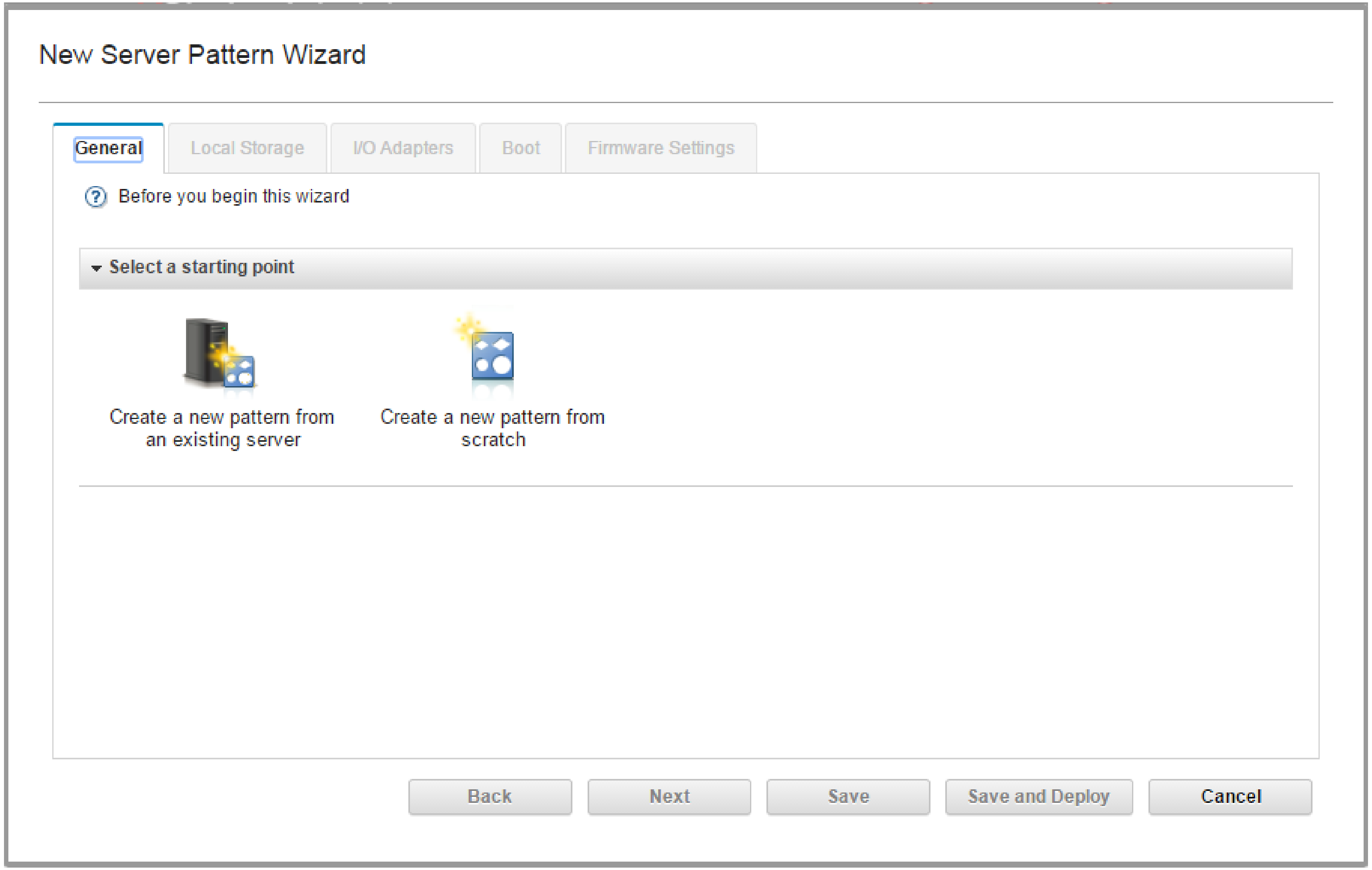
To begin using Configuration Patterns, either select a view from the Provisioning menu in the menu bar or select a task from: Suggested Tasks



lenovo

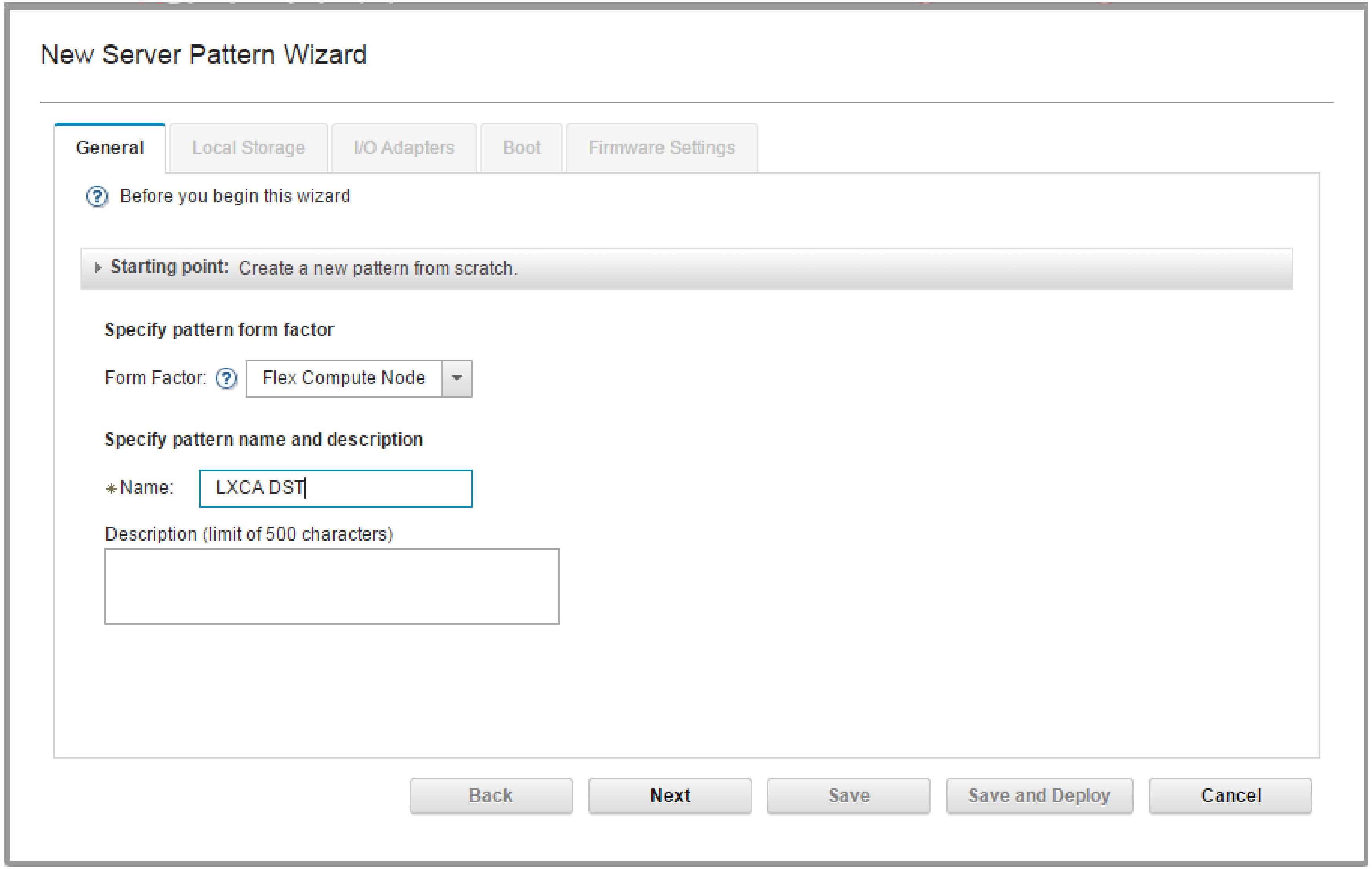
Configuration Patterns: New Server Pattern Wizard

- User starts by selecting either create from scratch or from existing server



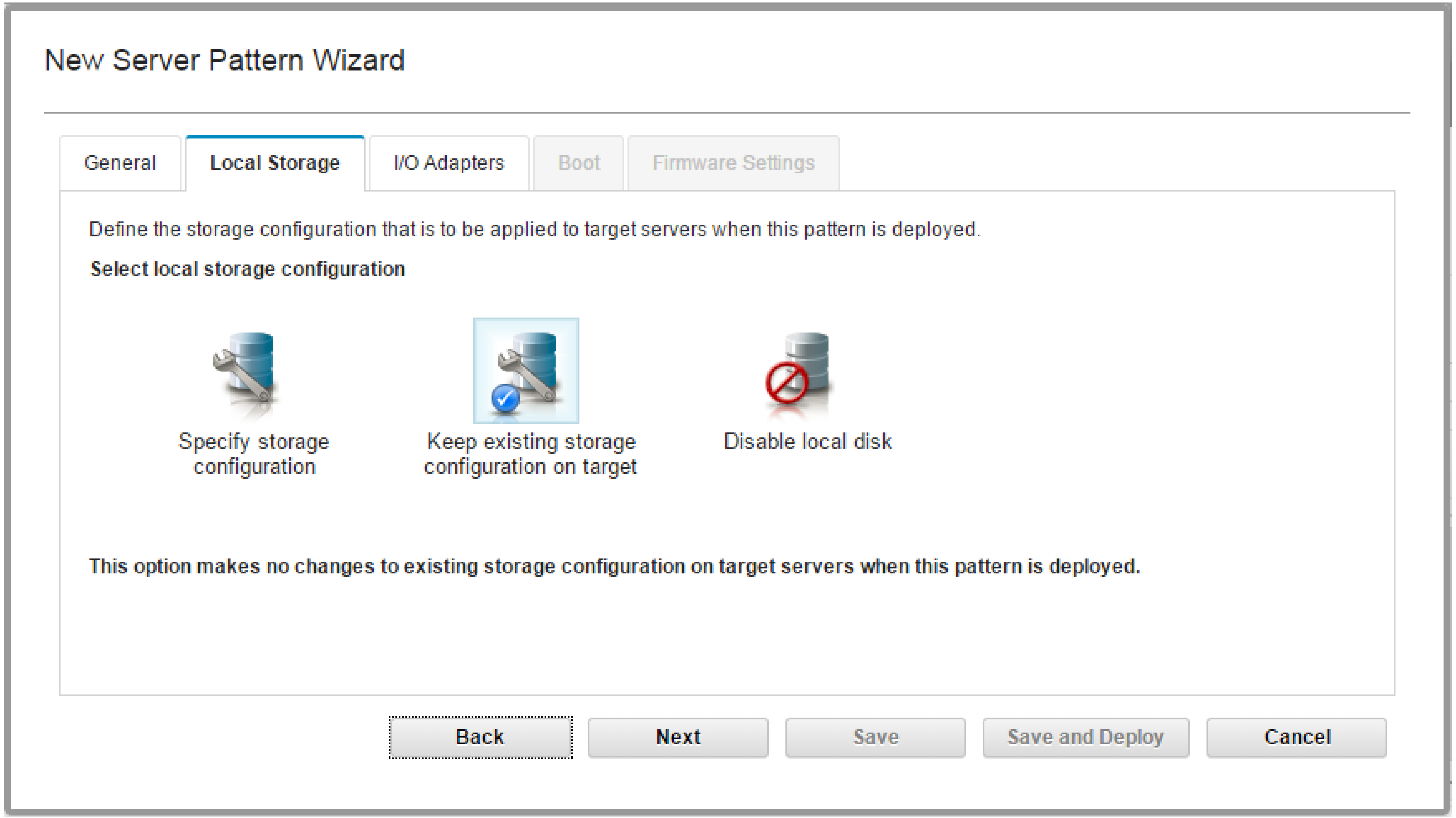
Configuration Patterns: New Server Pattern Wizard Step 1

- Select server type (Flex vs Rack) and enter pattern name



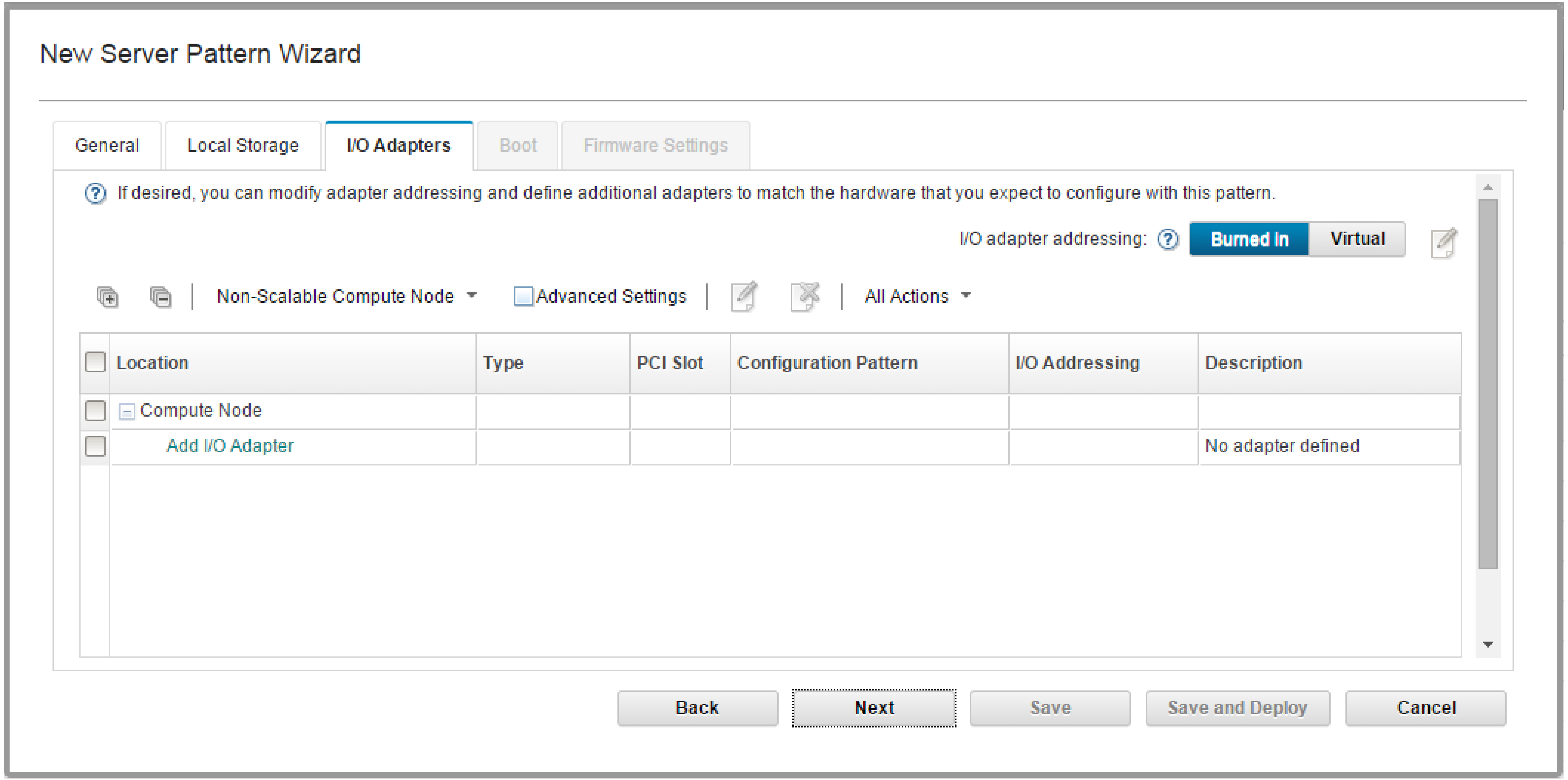
Configuration Patterns: New Server Pattern Wizard Step 2

- Local Storage: Basic RAID Boot volume, keep existing, or disable for SAN boot



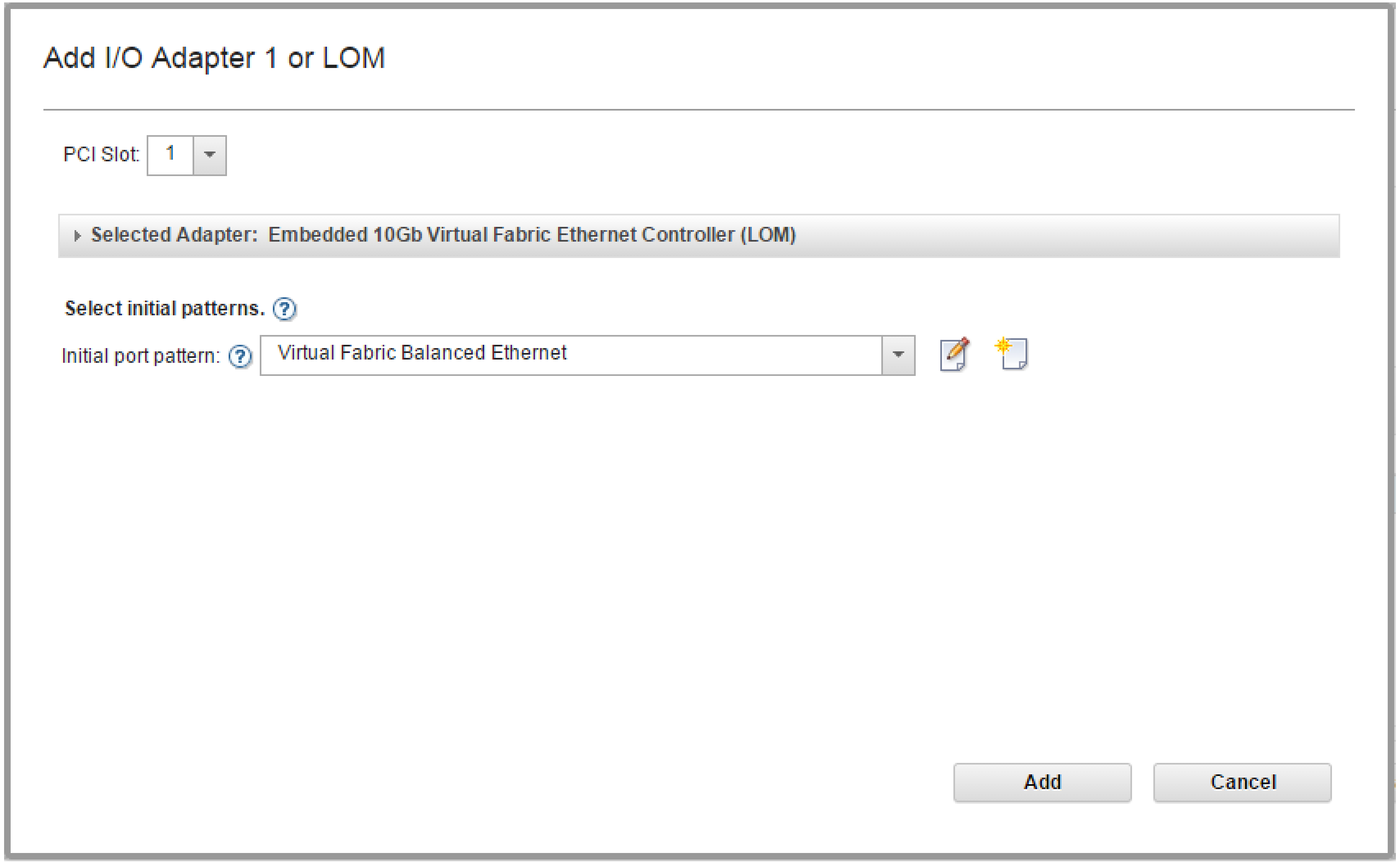
Configuration Patterns: New Server Pattern Wizard Step 3

- Define IO Adapter configuration



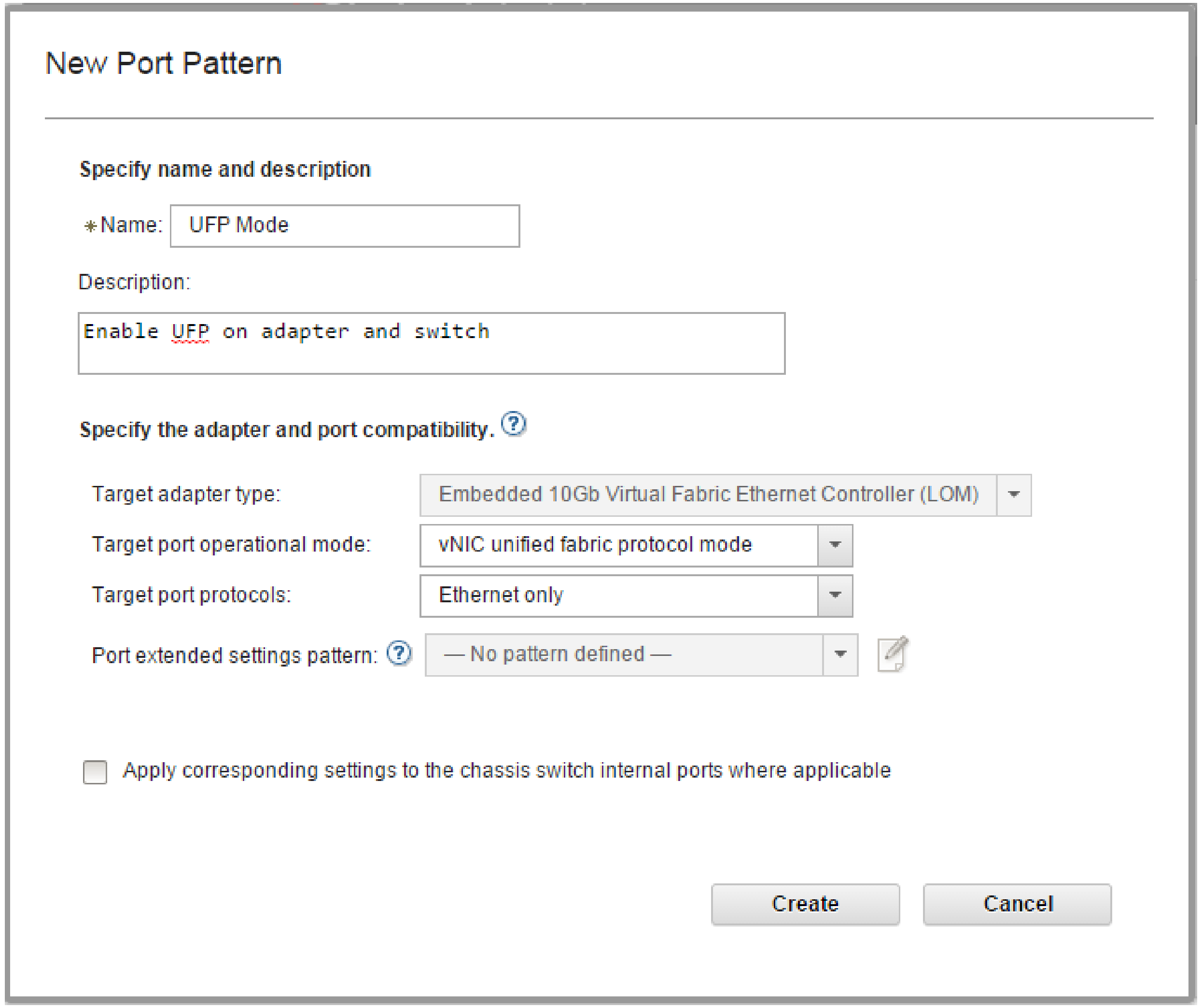
Configuration Patterns: IO Adapter

- Select IO Adapters and define settings via Port Pattern



Configuration Patterns: Port Pattern

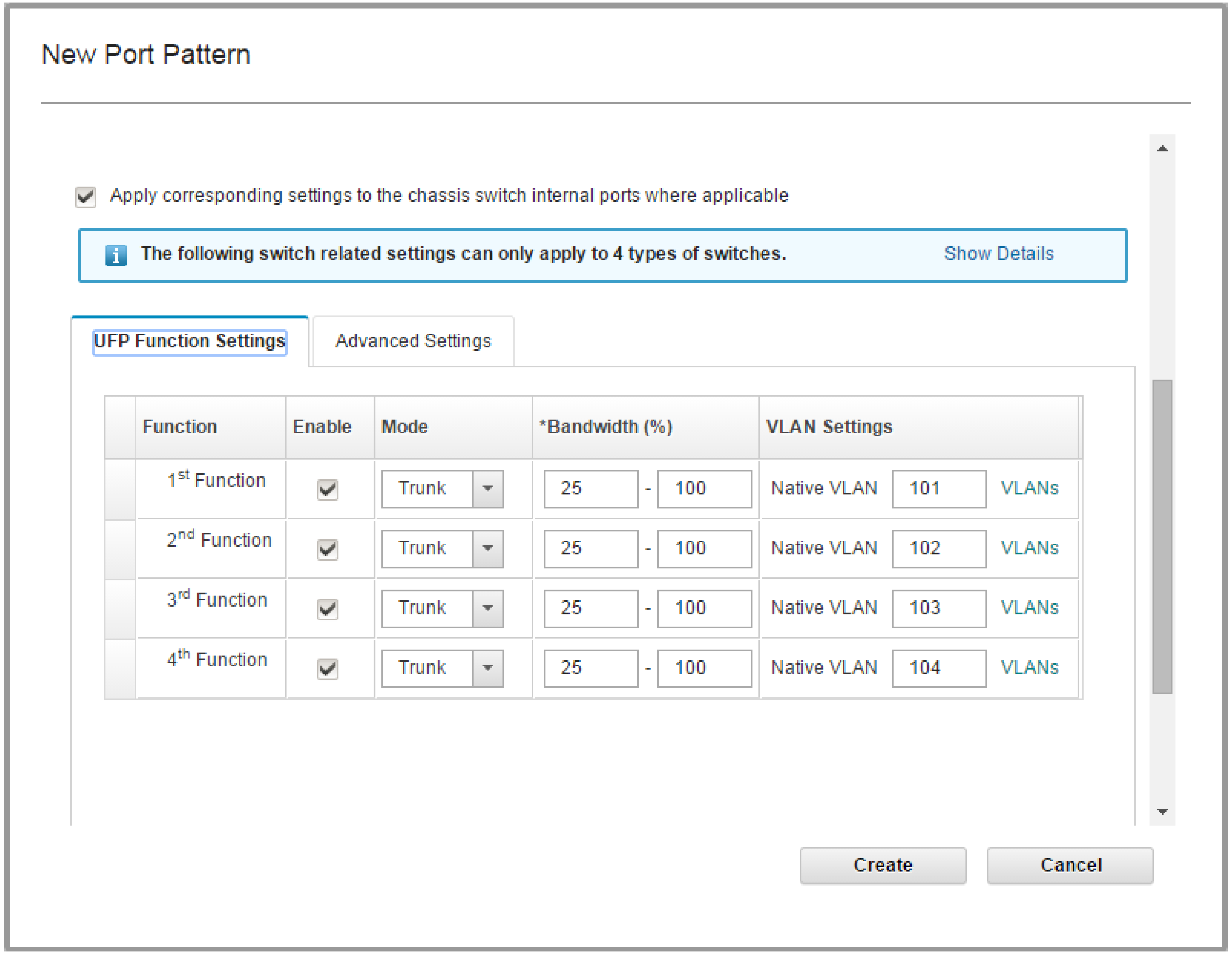
- Configure Ethernet & Fibre Port Settings



- Port Virtualization Settings
  - vNIC, UFP Mode
- Storage Protocols
  - FCoE, iSCSI enablement
- Extended Settings
  - Learned adapter settings for advanced Ethernet and Fibre Channel settings

Configuration Patterns: Port Pattern, Chassis Ethernet Switch Support

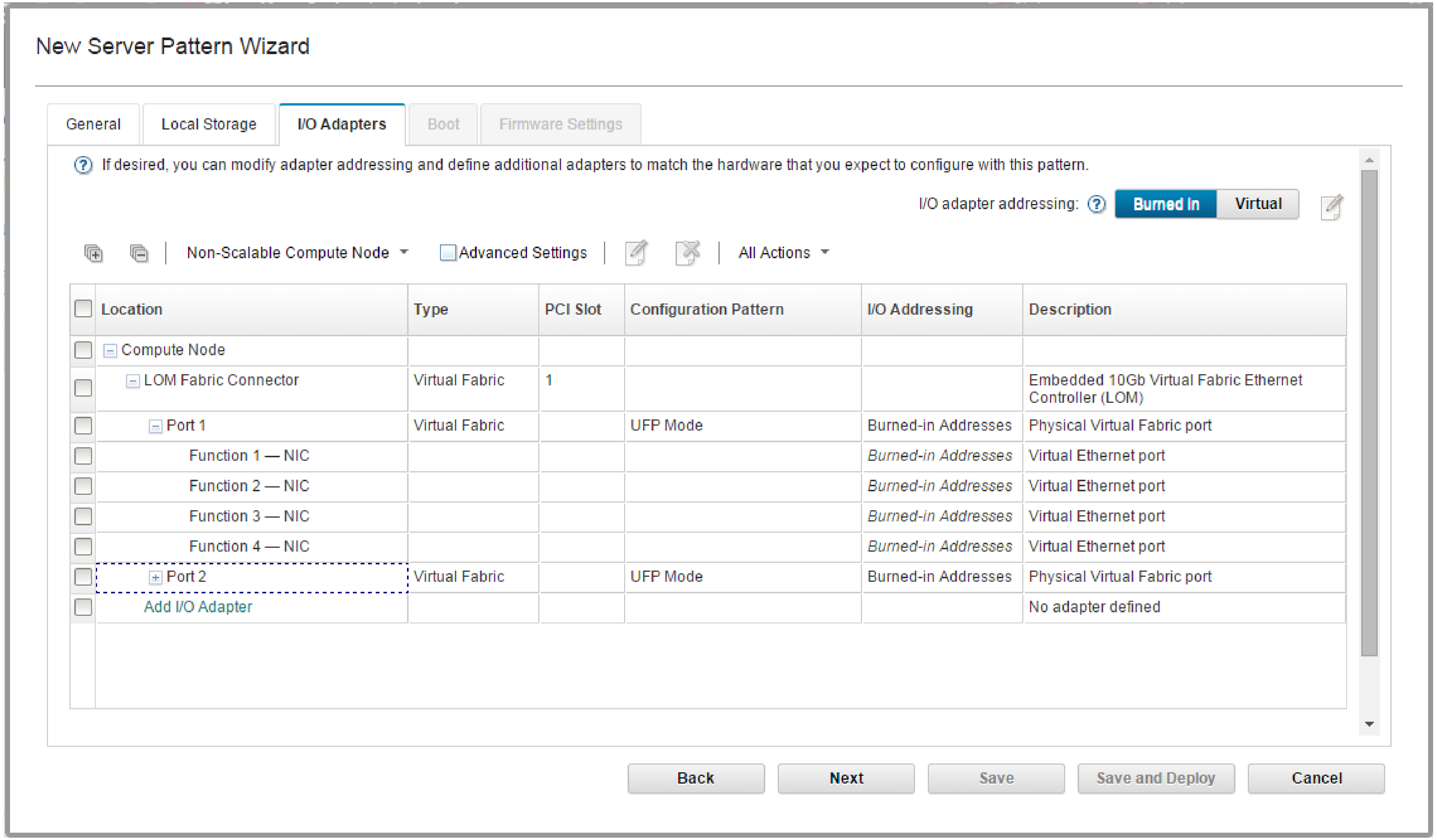
- New in LXCA, Support for configuring chassis Ethernet switch internal port settings



- Supported on Lenovo branded chassis switches:
  - EN4093R, CN4093, SI4093 & SI4091
  - Integrates configuration that requires coordination node and chassis switch configuration
  - Including, UFP Port Provisioning, VLAN Membership, flow control and failover

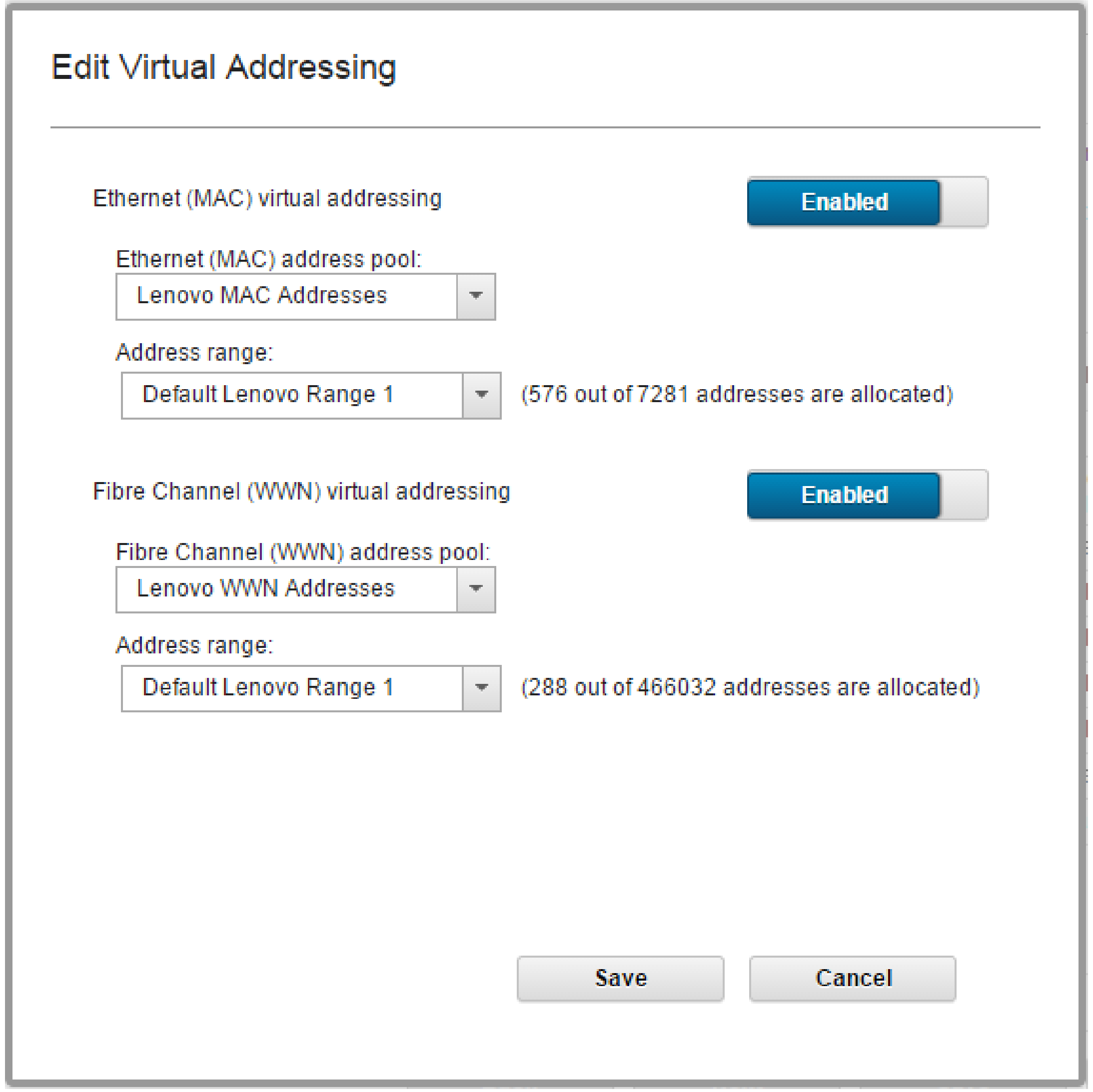
Configuration Patterns: IO Adapter View

- Adapter, port patterns and enabled ports displayed



Configuration Patterns: Address Virtualization

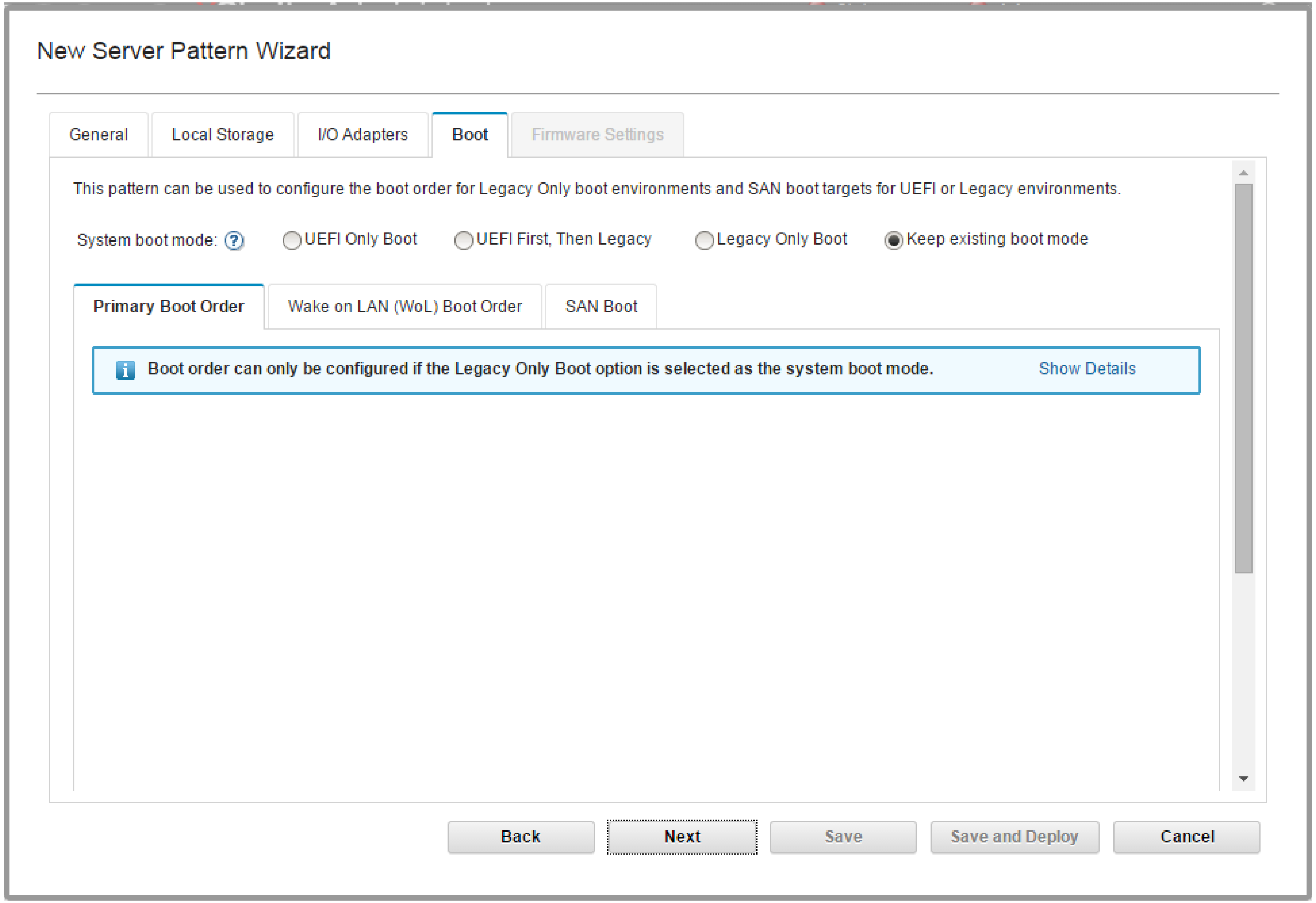
- Enable Virtualization of IO Hardware Addresses



- Virtual addressing
  - Provides ability to re-assign or “virtualize” IO Adapter Ethernet and Fibre port addresses, overriding burned in addresses
- Enables SAN Boot manual failover
  - By assigning addresses to a profile instead of burned in, profiles can be deactivated and activated to a new node upon node failure
- New in LXCA, support for defining custom address pools
  - Users can now define custom addresses pool sizes & starting addresses

Configuration Patterns: New Server Pattern Wizard Step 4

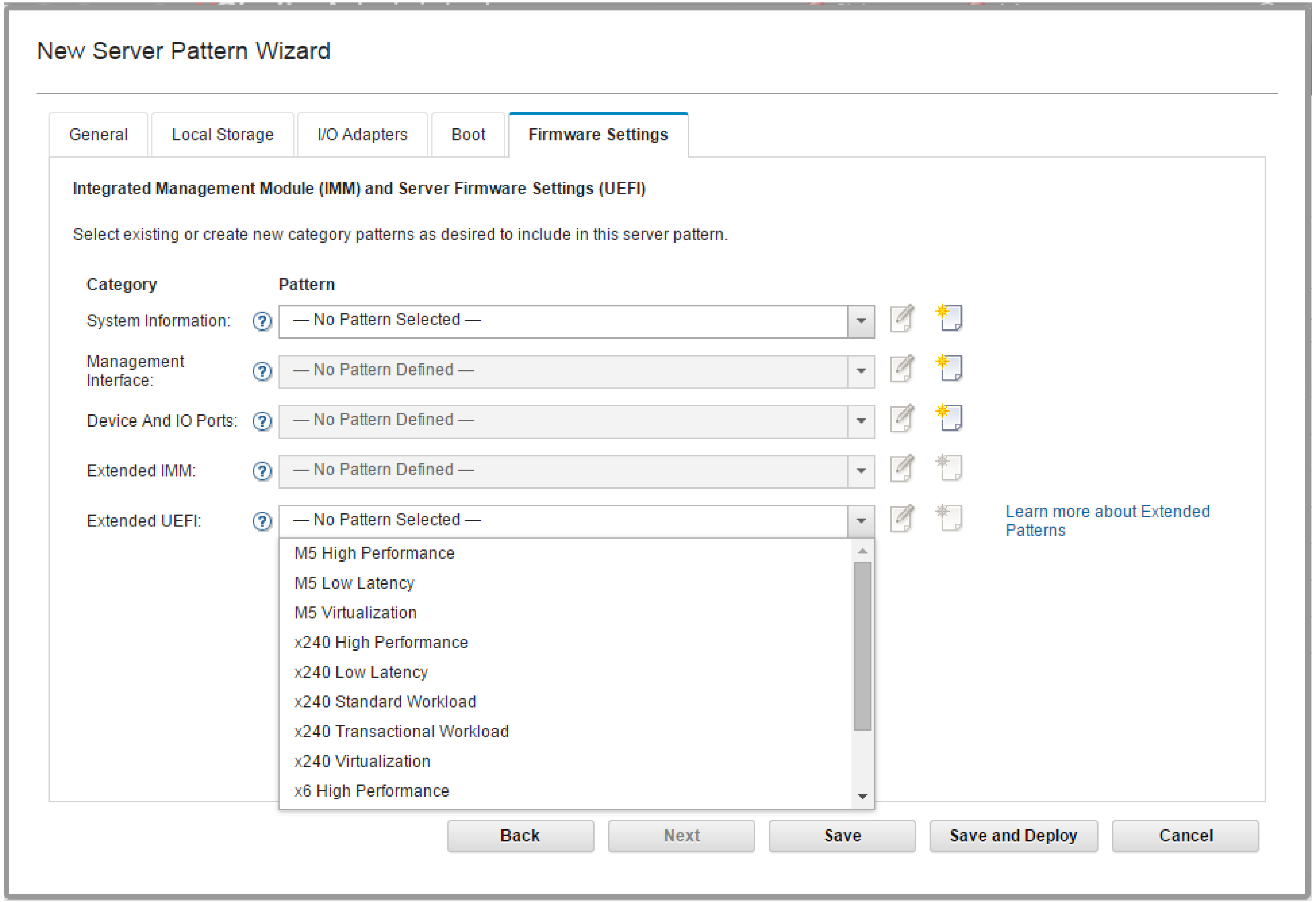
- Boot order



- Most useful for legacy boot
  - UEFI Oses typically modify the boot order as part of install
  - Defining boot order most useful for legacy boot installations

Configuration Patterns: New Server Pattern Wizard Step 5

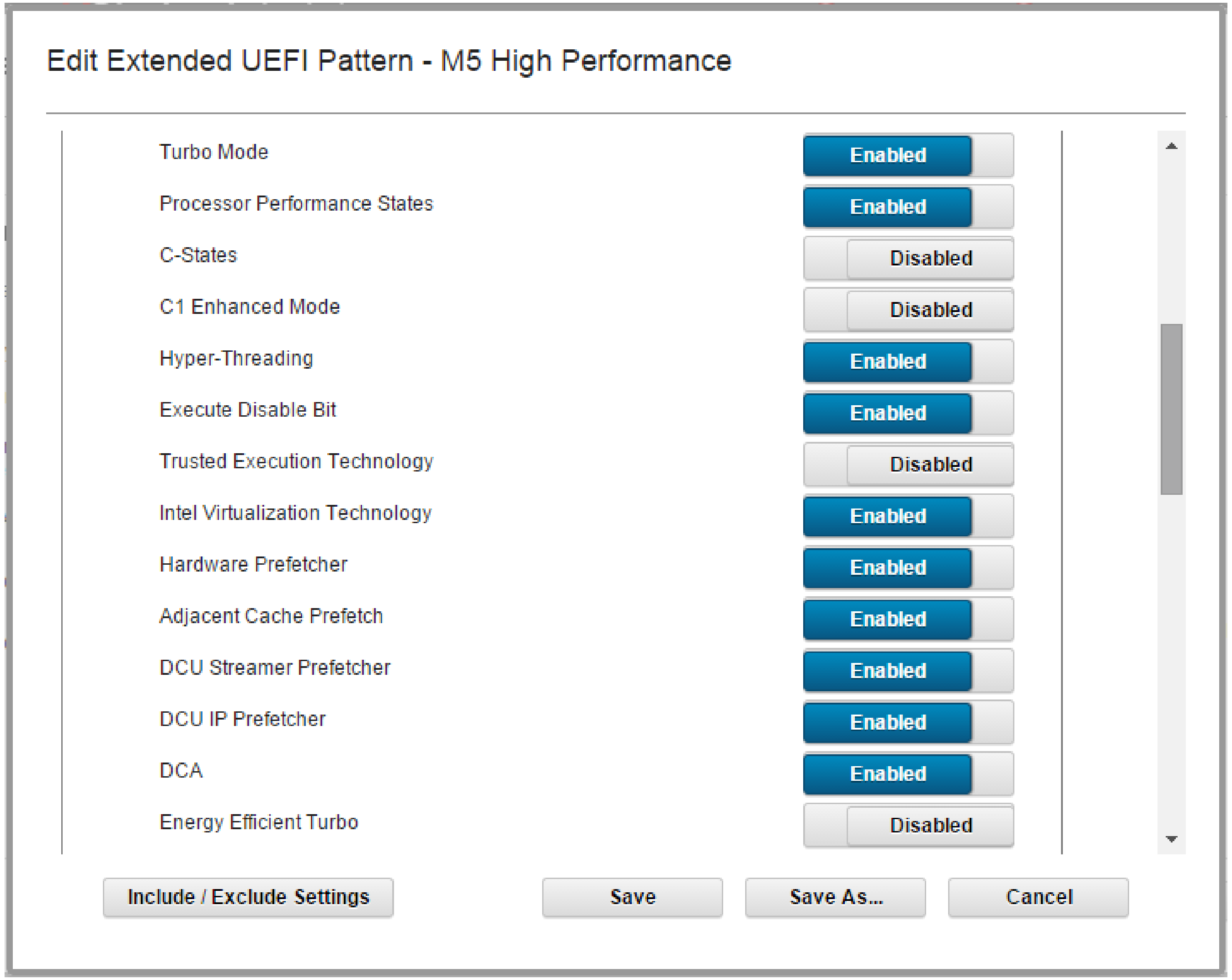
- Firmware Settings



- Patterns for modifying:
  - System info, including location & device name
  - Management Interface, IMM IP, Hostname
  - Device & IO Ports, Console redirection & SOL
  - Extended IMM & UEFI Settings

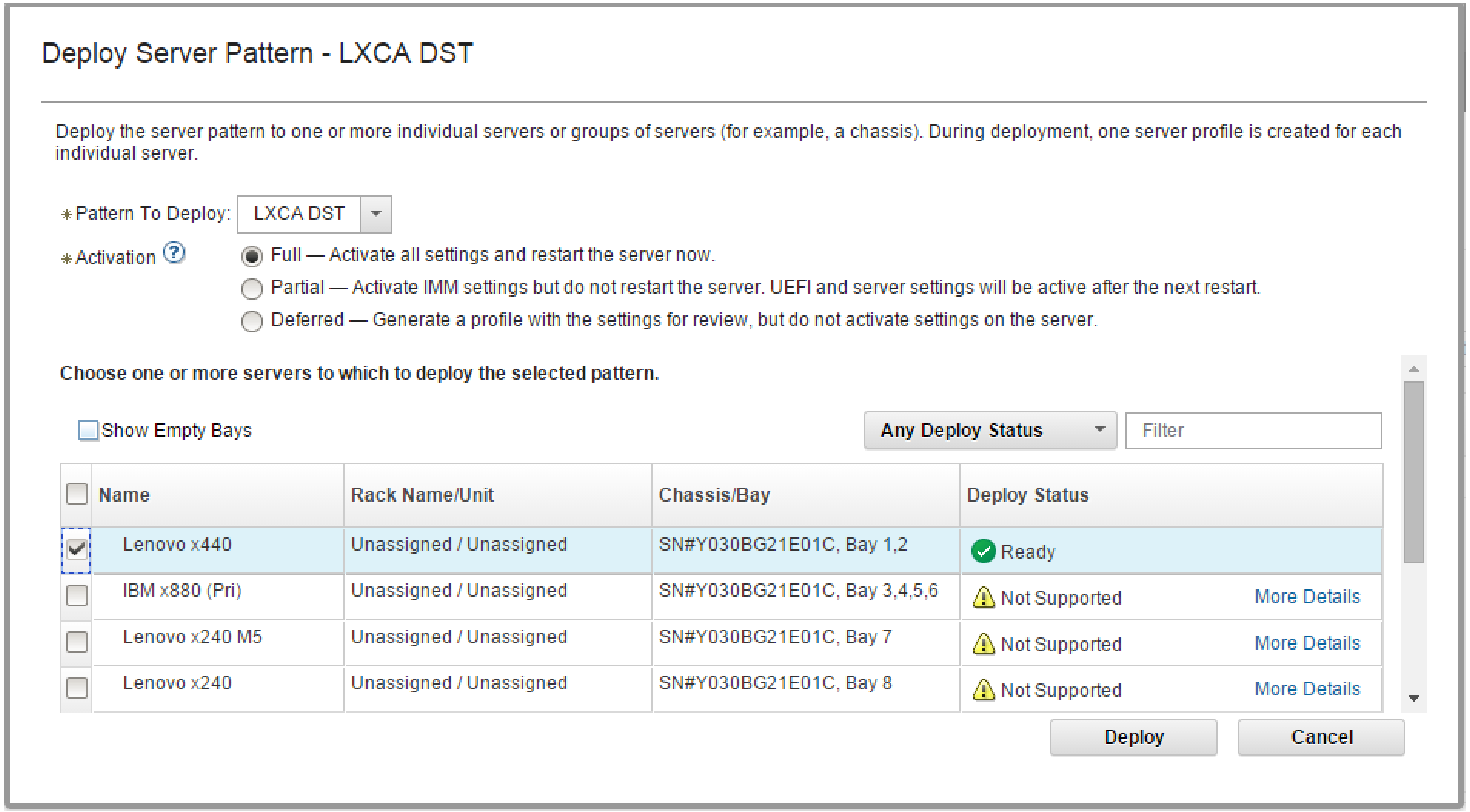
Configuration Patterns: Extended UEFI Patterns

- New in LXCA, Pre-defined UEFI Patterns



- Pre-defined, ATS provided best practices, optimized UEFI settings for typical workloads
  - Virtualization (hypervisors)
  - High Performance
  - Low Latency
  - Transactional workloads

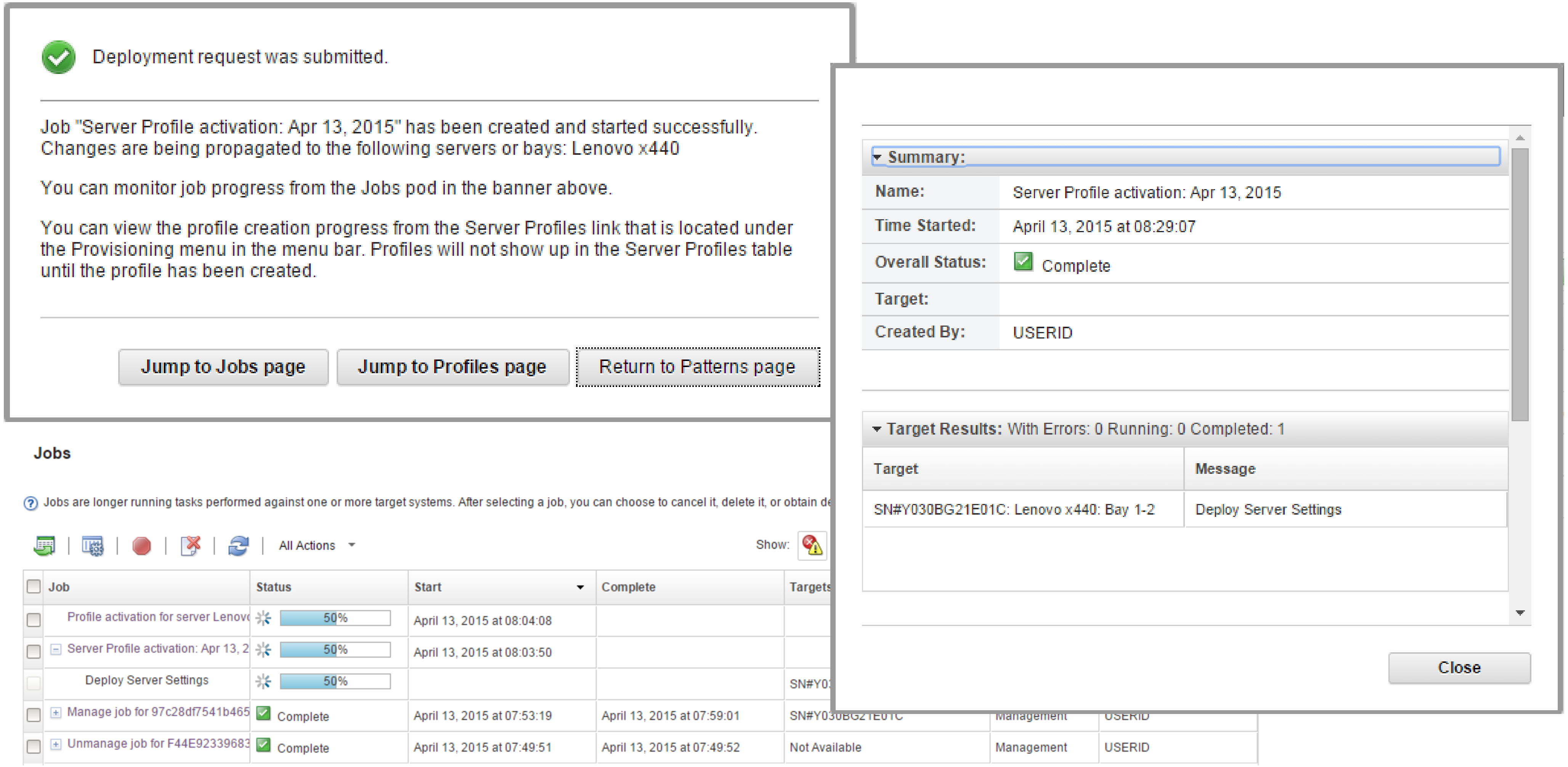
Configuration Patterns: Deploy Server Compatibility Verification



- Ensures server has minimum required firmware levels
- Server is available and not locked by another provisioning activity
- Validates selected Adapter Port Pattern is compatible with server

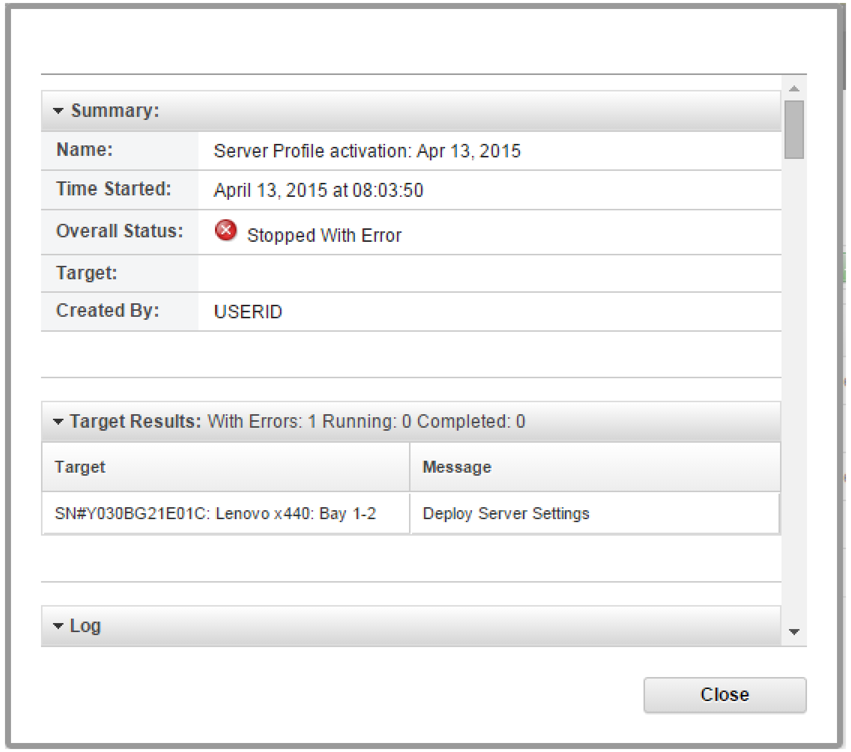
Configuration Patterns: Job Status

- Monitor deploy process via Jobs UI



lenovo

Configuration Patterns: Troubleshooting



- Most failures are captured and logged via the job log
  - Errors communicating with server
  - Failure to restart server
  - Incompatible settings
- Detailed failure data captured via FFDC captures
  - Capture ID: 5101
  - Includes logs, & diagnostic data
  - Archive of persisted user data, including created patterns, profiles & pools